

TrES-3b

R-2312

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120502 · created 2026-08-18T00:08:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456049.83977 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.015
Transit depth (Rp/Rs)^2	3.31 +/- 0.54 [%]
Orbital Inclination (inc)	82.98 +/- 0.68
Airmass coefficient 1 (a1)	1.024 +/- 0.013
Airmass coefficient 2 (a2)	-0.02 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#1 - [232, 43]
Optimal Aperture	3.32
Optimal Annulus	10.33
Transit Duration (day)	0.0658 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.015 vs 0.16309 (deviation 1.26σ)
Duration fit vs published	0.0658 d vs 0.0567 d (deviation 1.86σ)
Mid-time O–C	0.5 min
Depth SNR	12.1
Residual scatter	1.3 %

Light curve

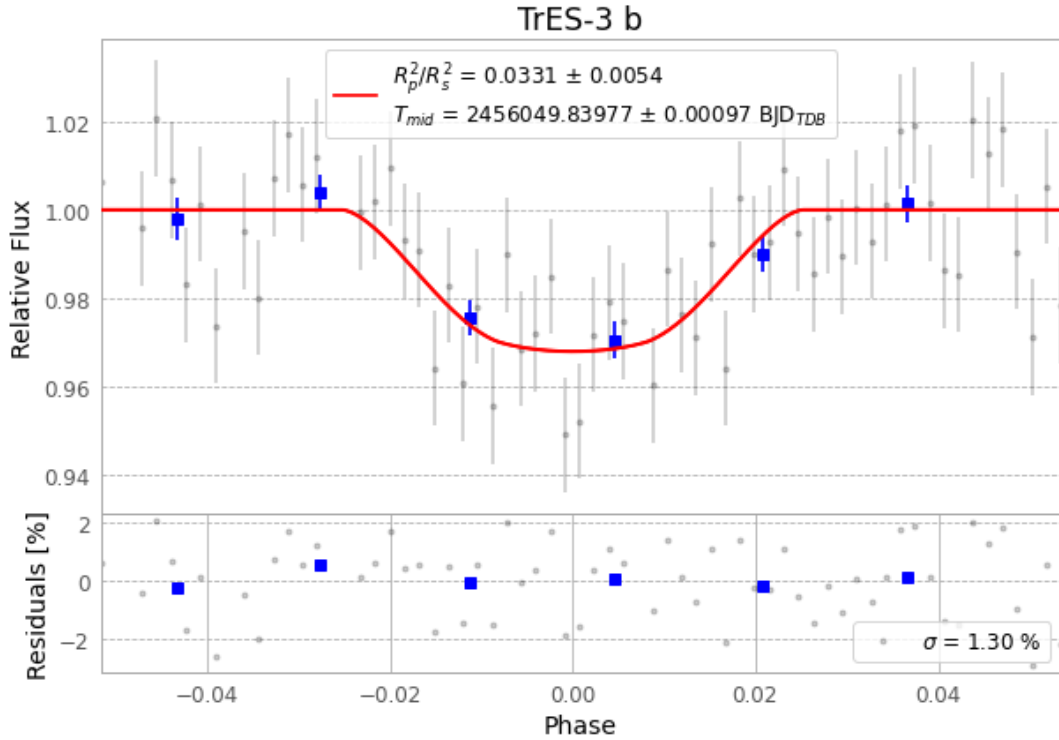


Figure 1 — FinalLightCurve_TrES-3 b_2012-05-01.png (SHA-256 413674568316d699...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a4d7ca369ed2991b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

63 FITS frames (41 MB), TRES-3120502062743.FITS → TRES-3120502094513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120502.log (724955398348...), FinalLightCurve_TrES-3 b_2012-05-01.png (413674568316...), FinalLightCurve_TrES-3 b_2012-05-01.csv (92ff03d67b47...)

Compute resources

Wall time 133.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:08Z.